SUPERNOVA × SPACEPOINT · THEME 03 · SEAS OF DATA

Radar sees every ship. We find the ones that hide.



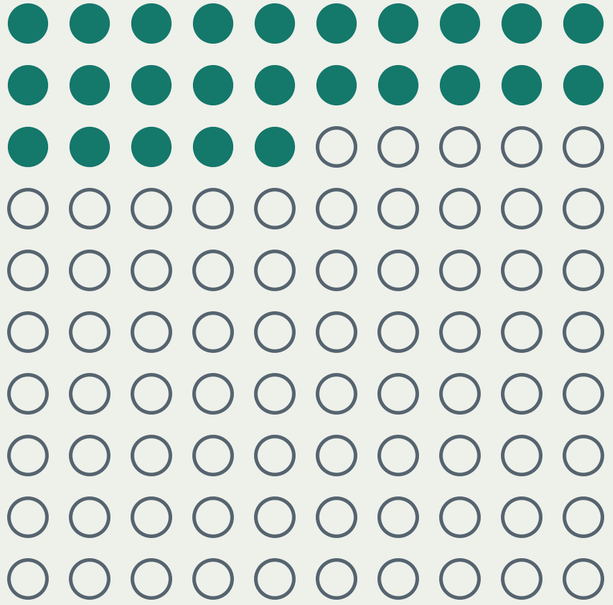
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BITS Pilani, Dubai Campus · November 2026

About three in four industrial fishing vessels are missing from public ship tracking

72–76%

of the world’s industrial fishing vessels are not publicly tracked. AIS is a signal each ship chooses to send.

Global figure, 2017 to 2021, not specific to the UAE. Dark does not mean illegal, and radar shows that a ship is there, not who it is.



Each circle = 1 of 100 vessels.
Filled = tracked.

SOURCE: PAOLO ET AL., NATURE 625, 85-91 (2024)

Sentinel-1 passes over Jebel Ali about every three days

COVERAGE

250 km swath, 10 m pixels, VV band

CONDITIONS

Day, night and through cloud

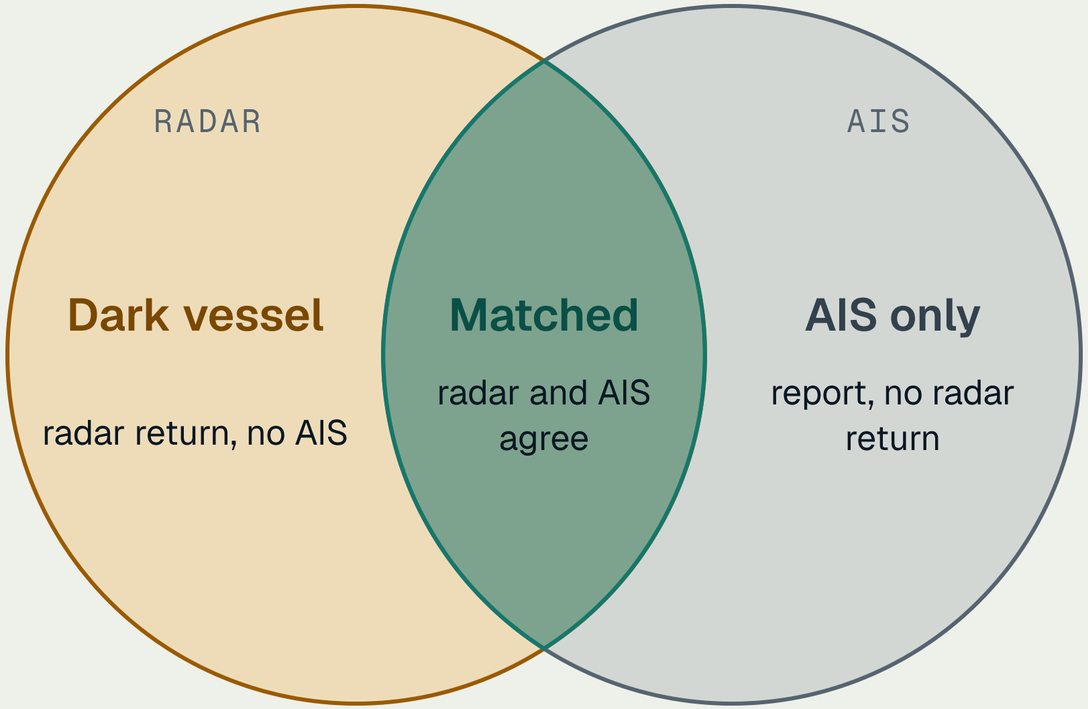
ACCESS

Free and open through Copernicus



PASSES FOUND IN THE COPERNICUS CATALOGUE · 25.01°N 55.06°E

Radar minus AIS equals the dark vessels



THE RULE

A ship is matched when an AIS report lies within 500 m of a radar detection. No report in range: it is flagged dark.

WHY IT HELPS

Dark is a lead, not a verdict. It tells patrols and port authorities where to look first.

IN THIS PROTOTYPE

AIS reports are simulated, because no free real Gulf feed exists. Every output is labelled that way.

One system: the satellite finds, the dashboard shows, the buoy confirms

01 · SPACE

Sentinel-1 radar

Free passes about every three days. Sees every hull, day, night and through cloud.

02 · PIPELINE

Detect and match

Sea mask, CFAR detector, map coordinates, then an AIS match within 500 m.

03 · DASHBOARD

See and act

Ships on a map, dark flags in amber, and a ship count for each port on every pass.

DARK-SHIP ALERT

AIS + PHOTO

THE LOOP

Radar flags a ship with no AIS report. The alert would go to the buoy nearest to it. The buoy's AIS log and camera confirm or clear the ship, and real AIS then replaces the simulated list.

04 · SEA

Solar buoy

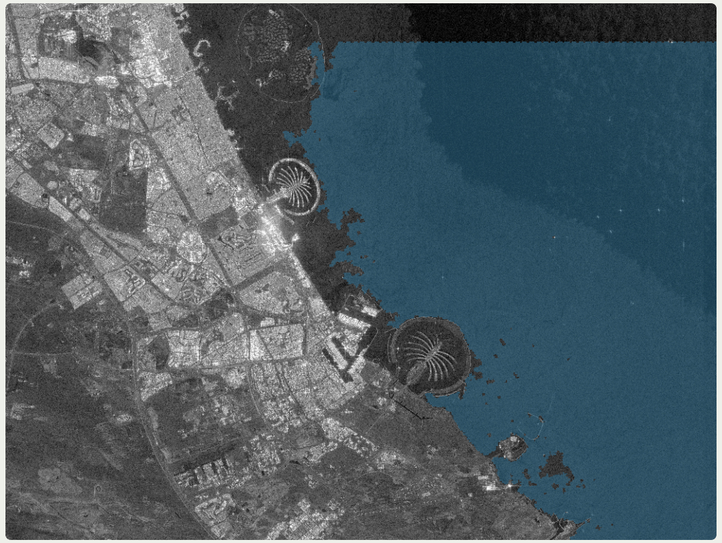
Listens for AIS, films the surface, senses oil. Design only, not built.

Radar in, ships out: three steps on one real pass over Dubai



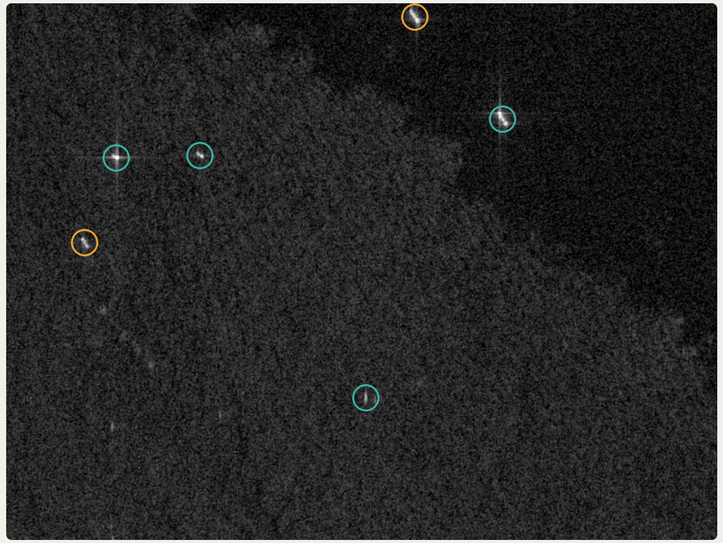
1 · RAW SCENE

2 Nov 2026, 02:14 UTC. Bright is land and hulls, dark is calm sea.



2 · SEA MASK

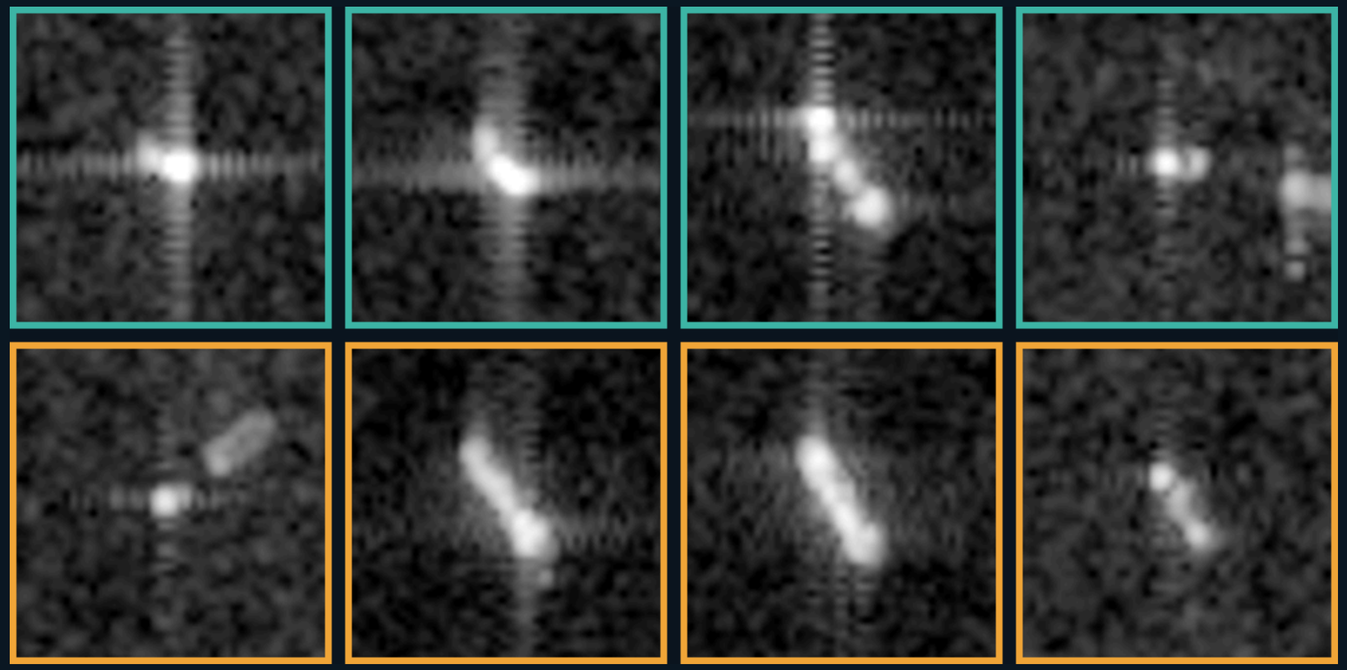
Blue is sea. The coast is mapped once from a calm pass and reused on every scene.



3 · DETECTIONS

CFAR keeps points far brighter than the sea around them. Teal: AIS match. Amber: none.

Close up, each ship is a bright streak; the amber ones have no AIS report



WHAT YOU SEE

A 480 m square around each detection. The cross is radar sidelobes, not part of the ship.

TOP ROW · MATCHED

An AIS report lies within 500 m.

BOTTOM ROW · DARK

No report in range. A lead for the buoy or a patrol to check.

Four real Gulf passes: 250 ships located and checked against a simulated AIS list



2 NOV 2026 · 02:14 UTC · 20 OF 73 SHIPS SHOWN

250
SHIPS

205
MATCHED

45
DARK

AIS IS SIMULATED

The dark share follows from how we simulate AIS, so it shows the method working, not a finding about Gulf shipping.

HOW TO READ IT

Teal ring: an AIS report within 500 m. Amber ring: none. Each amber ring is a lead to check, not an accusation.

One screen shows every ship, flags the dark ones and counts ships per port



THE MAP

All 250 detections from four passes. Red: no simulated AIS match. Blue: matched.

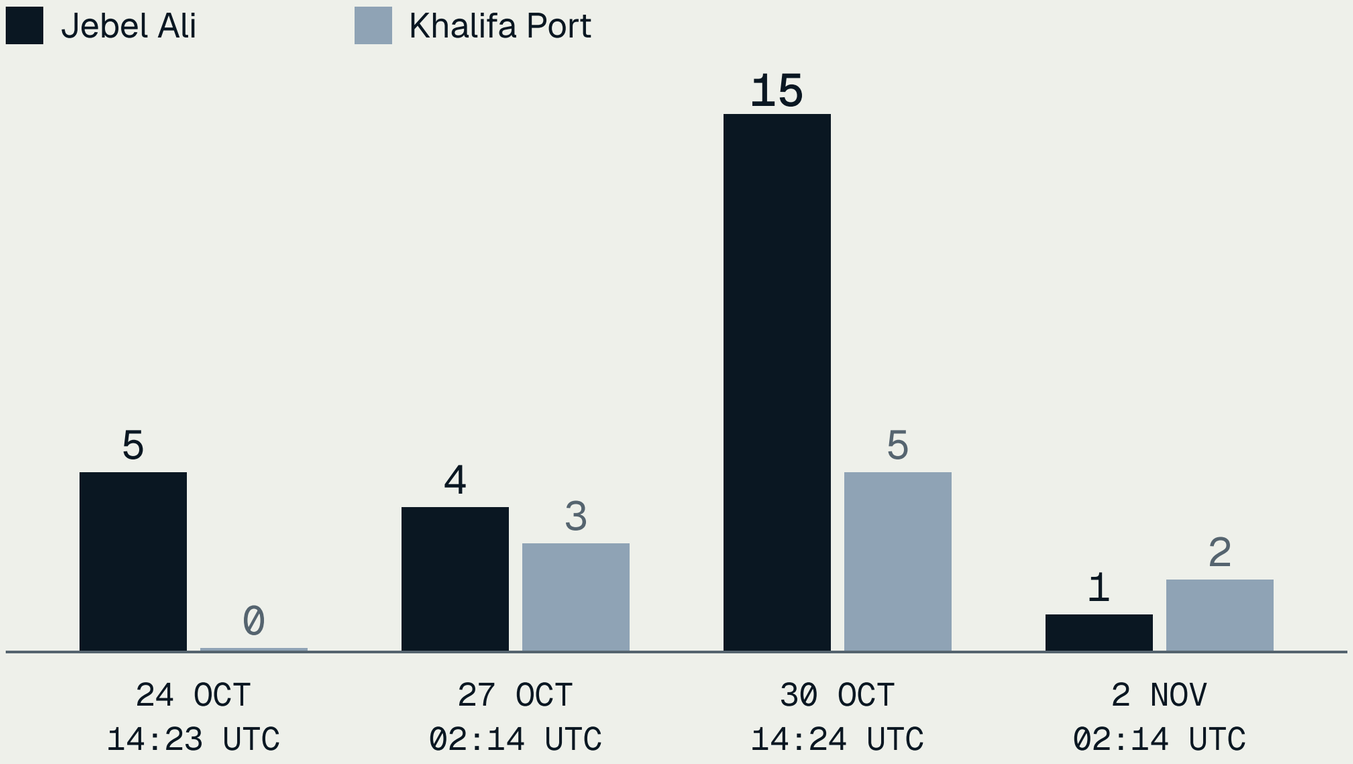
THE BANNER

A banner and every label say AIS is simulated, so the dark share is not read as a finding.

THE PORT PANEL

Raw counts per pass beside the index, with no trend line.

Four passes give a baseline, not a trend: Jebel Ali’s approaches held 1 to 15 ships



THE INDEX

Ships counted in each port box on a pass, divided by that port’s median across passes, times 100.

WHAT THE COUNTS ARE

Ships at anchor and in the approaches. Ships hidden among quay clutter are mostly not counted.

WHY NO TREND YET

Sea state and look angle change what radar picks up, and we have not corrected for them.

SHIPS DETECTED PER PORT BOX · 4 PASSES

On 1,962 radar images it never saw, the detector finds 82% of ships at 91% precision

5,604 HRSID IMAGES, SPLIT THREE WAYS



3,278 train

364 tune

1,962 never seen



Scores on the held-out test split. Bars run from 0 to 1.

WHY IT MATTERS

A dark flag is only as trustworthy as the detection under it. Misses hide ships and false alarms invent them.

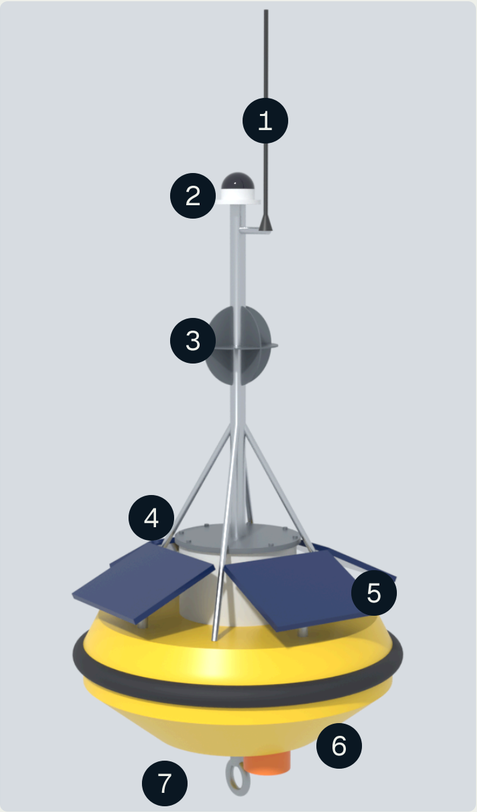
WHAT THIS IS NOT

HRSID pixels are finer than free Sentinel-1's 10 m, and the Gulf scenes have no labels, so Gulf accuracy is not measured.

WHAT WE DID ABOUT IT

The Gulf passes use a classical CFAR detector that needs no training. Appendix A2 has the settings.

A solar buoy would watch between satellite passes and check what radar flags



1 **AIS antenna**
Hears which ships are broadcasting.

2 **Camera dome**
Shows nearby ships, silent ones too.

3 **Radar reflector**
Makes the buoy easy to see on ships' radar.

4 **Electronics bay**
Sealed space for batteries, processor and radio link.

5 **Four solar panels**
Power everything, tilted outward to shed spray.

6 **Oil-sheen sensor**
Under the hull; flags possible oil in the water.

7 **Mooring eye**
Attaches the anchor chain on the centre line.

SIZE

1.8 m hull, about 4.5 m tall overall

THE CAD FILE

A STEP model with 14 named parts, built by a script from named dimensions.

DESIGN ONLY

Not built. No stability or mooring analysis yet; the waterline is a target, not a result.

What is simulated, what is assumed, and what comes next

Item	Status	Effect on the result
AIS reports	Simulated	Dark share is illustrative until a real feed is connected
Port capacity	Assumed	Scales berth occupancy only; the activity index is unaffected
Training data	Public HRSID	Finer pixels than Sentinel-1, so Gulf passes use a classical detector
Buoy	Design only	CAD model and renders, nothing built

NEXT

1. Connect a real AIS feed
2. Retrain on xView3-SAR, which labels dark vessels
3. Add oil-spill detection on the same radar scenes

Data, models and licences

Component	Used for	Source and terms
Sentinel-1 GRD	Real Gulf scenes, VV band	ESA, via Copernicus Data Space, free and open
HRSID	Training and held-out test of the ship detector	Wei et al., IEEE Access; repository under GPL-3.0
YOLO11	Detector, fine-tuned from pretrained weights	Ultralytics
Port boxes	Counting ships inside Jebel Ali and Khalifa Port	Drawn by us, checked against OpenStreetMap extents

Contains modified Copernicus Sentinel data 2026.

Method and parameters

Step	Setting	Reason
Sea mask	Static lon/lat grid, 0.001°, built from a calm-sea pass	Windy sea is as bright as desert, so brightness cannot find the coast
Detector	CFAR, 61 px window, 8 sigma, at least 3 px, at most 450 m	Needs no training and works at 10 m pixels
Merge	Returns within 150 m become one ship	One hull often splits into several blobs
Geocoding	Ground control points in each GRD file	Checked against Palm Jumeirah and Palm Jebel Ali
AIS match	Simulated: 80% reported, 120 m jitter; gate 500 m	No free real Gulf AIS feed
YOLO11s	Fine-tuned on HRSID, scored on 1,962 held-out images: precision 0.91, recall 0.82, mAP50 0.90	Benchmark score only; Gulf passes use CFAR